

Beyond Accuracy: An Analytical Comparison of Machine Learning Classifiers on Real-World Loan Approval Prediction Problem

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Abstract—Accurate loan status prediction is a critical task for financial institutions to mitigate risks effectively, especially as loan application volumes increase, necessitating reliable and automated models for faster, fairer and more equitable decision-making. The diversity of applicant profiles and the variability in data quality present serious obstacles and challenges to building universally effective prediction systems, complicated further by missing data and complex feature interactions. To address these issues, in this paper we propose an efficient framework combining advanced data preprocessing with machine learning algorithms such as Random Forest, Logistic Regression, Naive Bayes, SVC, Decision Tree, and K-Nearest Neighbors, evaluated on two large real-world datasets. These models, which receive training on diverse data demonstrated robust performance showing that for the first dataset, Random Forest gave an outstanding performance achieving the highest average accuracy of 92.33% and outperforming other algorithms in recall, precision, and F1-score. In contrast, Decision Tree achieved the best results on the second dataset with an impressive accuracy of 97.35%, surpassing competing models. As shown in this paper, combining advanced machine learning techniques with comprehensive data preprocessing can significantly improve the accuracy and reliability of loan status prediction. This approach offers a practical and effective solution for financial institutions to enhance credit risk management and make better lending decisions.

Keywords: Loan Status Prediction, Machine Learning, Logistic Regression, Naive Bayes, Decision Tree, Naive Bayes, Support Vector Classifier, K-Nearest Neighbors, Credit Risk Management, Automated Loan Approval, Predictive Modeling, Financial Institutions.

I. INTRODUCTION

Accurate loan status prediction plays a pivotal role in the financial sector, serving as a cornerstone for appropriate credit management and effective risk mitigation. Automated loan approval systems encouraged by advanced data-driven models offer the promise of not only speeding up decision-making but also enhancing fairness in lending practices. With millions of consumers relying on credit for essential

needs such as education, healthcare, housing, and business expansion, the ability to precisely evaluate creditworthiness directly impacts economic stability and individual financial health. Furthermore, efficient loan prediction systems help reduce default rates, improve portfolio quality, and optimize capital allocation, all of which contribute to the robustness of the broader financial ecosystem. Consequently, developing reliable and scalable loan prediction models is critical objective for banks and lending institutions striving to maintain competitiveness and regulatory compliance.

Due to the continuously growing volume of loan applications handled by financial institutions globally, relying on manual assessment methods has become increasingly inefficient and susceptible to bias. Accurately predicting loan outcomes remains a challenging task due to several underlying complexities. Applicants vary significantly in terms of age, income, employment background, and credit history, introducing high levels of heterogeneity. This complexity is further exacerbated by data quality issues, including missing entries, noise, and incomplete financial records. Additionally, the relationships between critical features—such as income consistency, requested loan amounts, and prior defaults—tend to be nonlinear and highly interdependent, rendering conventional rule-based evaluation models insufficient.

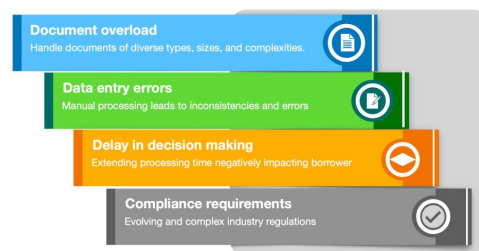


Fig. 1. Key Challenges in Manual Loan Processing

These challenges requires the implementation of advanced computational approaches that can process high-dimensional, large-scale datasets and identify complex patterns linked to loan repayment behavior. Furthermore, compliance with regulatory standards and the imperative to ensure fairness in lending introduce additional design constraints, making it essential for predictive models to be both interpretable and transparent, in addition to being accurate.

To address these challenges, this paper employs machine learning techniques to develop an efficient loan prediction framework. Machine learning, a subset of artificial intelligence, enables computers to learn from data, making it well-suited for predictive analytics in finance.

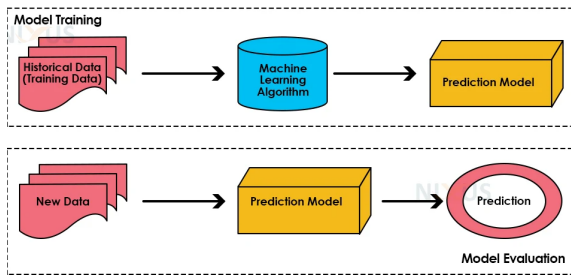


Fig. 2. illustration of Machine learning models

In this work, Supervised learning is used to train models on labeled loan data, including financial ratios and borrower details. Data preprocessing ensures reliability through handling missing values, encoding, and normalization. Multiple models, like Random Forest, Logistic Regression, Decision Trees, SVM, Naive Bayes, and KNN, are tested for loan status prediction.

The main contribution of this paper can be summarized as follows:

- Developing an efficient loan prediction framework
- Combining advanced data pre-processing with multiple machine learning algorithms.
- Evaluating and comparing the performance of diverse classifiers including Random Forest, Decision Tree, Logistic Regression, SVM, Naive Bayes, and KNN on large, real-world datasets.
- Demonstrating the effectiveness of machine learning in handling complex, heterogeneous data with missing values and nonlinear feature interactions.

The rest of the paper is organized as follows: Section II reviews related work. Section III describes the proposed methodology. Section IV Results and Analysis. Finally, Section V Concludes the study and suggests future research directions.

II. RELATED WORK

The authors discovered [1].his study evaluates deep learning models (MLP, CNN, LSTM, ResNet, DenseNet) and conventional classifiers (Random Forest, Logistic Regression,

Decision Trees) on a dataset of 252,000 records. SMOTE and SMOTE-TOMEK were used to address class imbalance, with DenseNet performing the best.

A study [2] uses machine learning to predict loan approval with a dataset of 1,000 applications. Nine classifiers were assessed, with SVM achieving the best performance: 90.55

The study in [3] explores machine learning models for automating bank loan decisions, including logistic regression, decision trees, random forests, SVM, and GBM. Using a large loan dataset, GBM outperformed others with 89.5% accuracy and 90.5% F1-score, proving valuable for financial decision-making.

Machine learning is not limited to education but has shown transformative potential across multiple domains such as healthcare, natural language processing, finance, and many other fields. Its ability to analyze complex datasets, uncover hidden patterns, and generate accurate predictions makes it a versatile tool for solving real-world challenges. The adaptability of machine learning methods continues to drive innovation and create opportunities for impactful applications across industries [4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]

III. PROPOSED METHODOLOGY

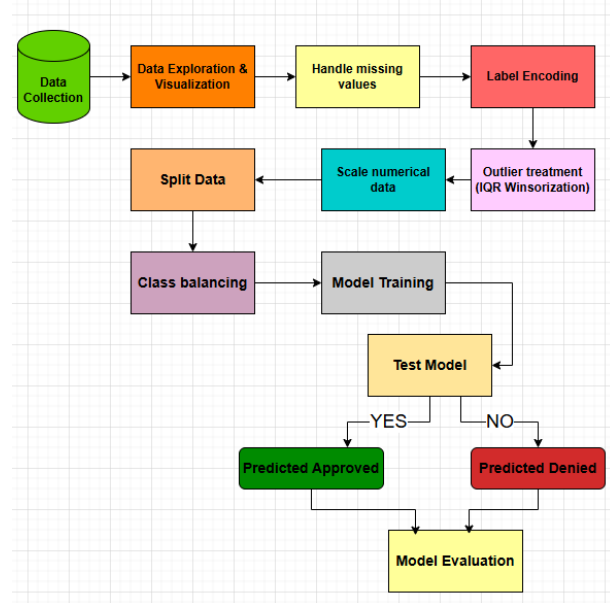


Fig. 3. Workflow of the Loan Approval Prediction System

A. Datasets Descriptions

1)Dataset 1:

A loan dataset of 45,000 records and 14 features shown in table I spans a diverse range of loan applications, providing a strong foundation for predictive modeling aimed at assessing loan approval outcomes.

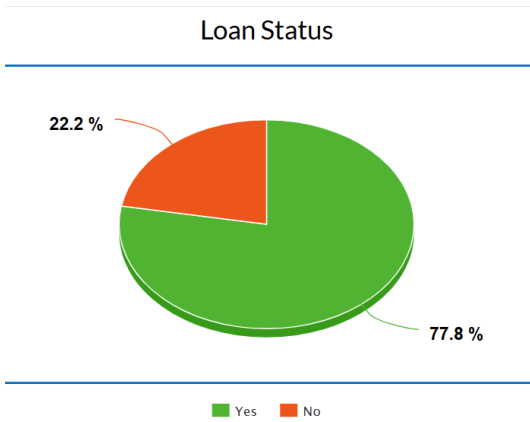


Fig. 4. Dataset Bank Loan pie chart

The above pie chart of figure 4 shows Distribution of loan status in the dataset, where approximately 77.8% of the records correspond to class 0 (e.g., loan rejected) and 22.2% correspond to class 1 (e.g., loan approved).

TABLE I
FEATURES OF LOAN_DATA.CSV

Features	Type	Description
Loan ID	Categorical	ID
Gender	Categorical	Gender of the applicant
Married	Categorical	Marital status
Dependents	Categorical	Number of dependents
Education	Categorical	Education level
Self_Employed	Categorical	applicant is self-employed or not
ApplicantIncome	Numerical	Income
CoapplicantIncome	Numerical	Income of the co-applicant
LoanAmount	Numerical	Amount requested
Loan_Amount_Term	Numerical	duration of the loan in months
Credit_History	Categorical	applicant has credit history or not
Property_Area	Categorical	Location
Loan_Status	Categorical	Target variable

The second and last loan dataset was designed for binary classification tasks with 4,269 data rows and 13 features as shown in table II captures detailed borrower and financial information relevant to loan approval decisions with a target variable indicating loan approval status. Approximately 62.2% of the loans are approved, while 37.8% are rejected.

TABLE II
FEATURES OF LOAN_APPROVAL_DATASET.CSV

Features	Type	Description
loan_id	Categorical	ID
no_of_dependents	Categorical	Number of dependents
education	Categorical	Education level
self_employed	Categorical	Yes or No
income_annum	Numerical	Annual income
loan_amount	Numerical	Amount of loan requested
loan_term	Numerical	Repayment period in months
cibil_score	Numerical	Credit score
residential_assets_value	Numerical	Value of residential assets
commercial_assets_value	Numerical	Value of commercial assets
luxury_assets_value	Numerical	Value of luxury assets owned
bank_asset_value	Numerical	Value of bank assets held
loan_status	Categorical	target variable

B. Data Preprocessing

1) Dataset 1: Preprocessing Steps:

1) Data Loading:

The first step involved loading the Loan Prediction (1) dataset, which was stored in a CSV format.

2) **Data Transformation:** features ('Gender', 'Married', 'Education', 'Self_Employed', 'Property_Area') were label-encoded, and missing numerical values (e.g., 'LoanAmount') were imputed using medians to standardize data and improve model robustness.

3) Handling Missing Values:

The `isnull().sum()` function confirmed no missing values in the dataset.

4) **Dealing with outliers:** were handled using IQR-based winsorization on columns like `loan_amt`, `loan_int_rate`, and `person_income`, reducing outliers significantly, e.g., from 2188 to 0 in `person_income`.

5) **Data Balancing:** The target variable (`loan_status`) had a class imbalance. Data balancing techniques equalized the class distribution, helping the model learn both outcomes effectively for better predictions.

2) Dataset 2: Preprocessing Steps:

1) Data Loading:

The Loan Prediction (2) dataset, stored in CSV format, was first loaded.

2) Handling Missing Values:

The `isnull().sum()` function confirmed no missing values, eliminating the need for imputation.

3) **Dealing with outliers:** The IQR approach showed no significant outliers in numerical features, maintaining dataset integrity.

4) **Encoding & Standardization:** `LabelEncoder` was used to convert categorical target variables (`loan_status`) into numeric form, and `StandardScaler` standardized the input features, ensuring a mean of zero and a standard deviation of one.

5) **Data Balancing:** Data balancing addressed class imbalance in `loan_status`, ensuring equal class representation for improved prediction accuracy.

C. Data Visualization

Data visualization is crucial in data science as it reveals patterns, trends, and outliers that are hard to see in tables, enhancing understanding, decision-making, and predictions.

1) Dataset 1 Visualization:

1) Loan Amount Distribution:

A histogram of the loan_amnt To evaluate the loan amounts in the dataset, a column was plotted. This graph displayed the loan sizes' frequency distribution. 5, which indicated that the majority of loans are between \$5,000 and \$15,000. This information is crucial for evaluating the risk profile of typical loan applications.

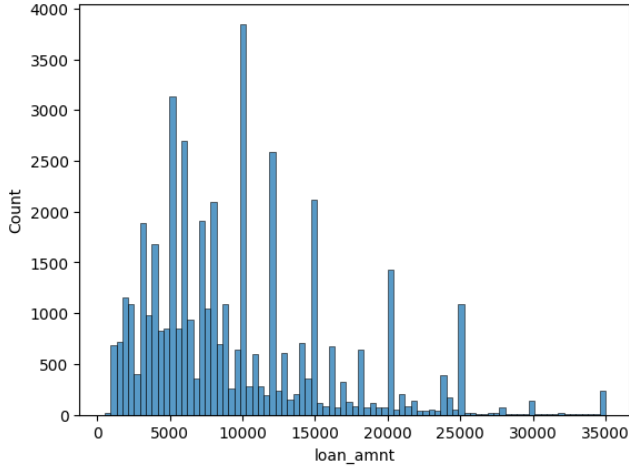


Fig. 5. Distribution of Loan Amounts

2) Dataset 2 Visualization:

a) Average Monthly Sales :

As shown in Figure 6 The histogram shows the loan term distribution, with the y-axis representing the number of loans and the x-axis showing loan duration. Most loans fall between the 5–10 year range.

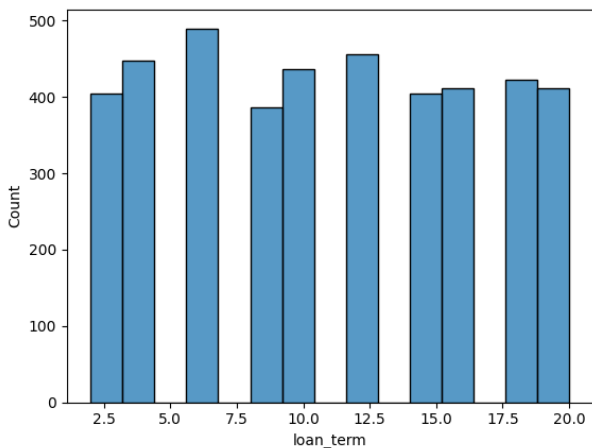


Fig. 6. Monthly Sales

b) Effect of Data Balancing on Dataset Distribution:

The graph in Fig 7 The right plot shows the

actual loan approval distribution, while the left shows balanced data, highlighting the importance of balance for machine learning models.

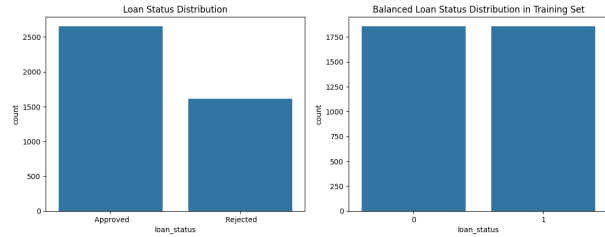


Fig. 7. Before and after Balancing

D. Used Algorithms

The models chosen were **KNN**, **Logistic Regression**, **Decision Trees**, **Random Forests**, **SVC**, and **Naive Bayes**. KNN and SVC handle decision boundaries, tree-based models capture complex patterns, while Naive Bayes and Logistic Regression provide quick baselines. Performance comparisons follow.

a) K-Nearest Neighbors (KNN):

KNN labels data by majority class. Small K is noisy, large K complicates decisions. It was slow and sensitive to outliers but worked well with scaled data.

$$D(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2} \quad (1)$$

$$\hat{y} = \text{mode}(y_i) \quad \text{for } i \in \mathcal{N}_k(x) \quad (2)$$

where $\mathcal{N}_k(x)$ denotes the k nearest neighbors of instance x .

b) Logistic Regression:

A sigmoid function in logistic regression predicts probabilities for binary classification, identifying linear trends in variables like income and occupation, making it a useful baseline for loan prediction.

$$P(y = 1 | \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}} \quad (3)$$

c) Decision Tree:

Decision trees classify data without feature scaling, capturing non-linear relationships and highlighting key variables like credit history and education. However, without pruning, they may overfit. **1. Gini Impurity**

$$G(t) = 1 - \sum_{i=1}^C p_i^2 \quad (4)$$

2. Entropy

$$H(t) = - \sum_{i=1}^C p_i \log_2(p_i) \quad (5)$$

3. Information Gain

$$IG(t, A) = H(t) - \sum_{v \in \text{Values}(A)} \frac{|t_v|}{|t|} H(t_v) \quad (6)$$

d) Random Forest:

Random Forest creates decision trees with random feature subsets, averaging predictions to boost accuracy and reduce overfitting, highlighting key variables like income and loan amount for reliable loan prediction.

$$\hat{y} = \text{mode}(h_1(\mathbf{x}), h_2(\mathbf{x}), \dots, h_T(\mathbf{x})) \quad (7)$$

where $h_i(x)$ is the prediction from the i -th decision tree.

e) Support Vector Classifier (SVC):

SVMs use kernels for non-linear data and optimal classification. SVC is computationally intensive but performs well on scaled data with linear/RBF kernels, though slower than Random Forest.

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t. } y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 - \xi_i, \xi_i \geq 0 \quad (8)$$

f) Naive Bayes:

Naive Bayes, based on Bayes' Theorem and feature independence, is efficient and fast, providing reliable results for scalable loan prediction with categorical variables like property area, gender, and education.

$$\hat{y} = \arg \max_{c \in C} P(c) \prod_{i=1}^n P(x_i | c) \quad (9)$$

E. Performance Metrics

Standards to measure classical models include accuracy (correct vs. total predictions), precision (true positives), recall (detected positives), and the F1-score (harmonic mean of precision and recall).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

• 5-Fold Cross-Validation

Table III shows 5-fold cross-validation results on Dataset 1. Average accuracies: KNN (0.8875), Logistic Regression (0.8903), Decision Tree (0.9159), Random Forest (0.9212), SVC (0.9055), Naive Bayes (0.7302). Random Forest had the highest accuracy, and Naive Bayes the lowest.

TABLE III
5-FOLD CROSS VALIDATION ACCURACY SCORES FOR DATASET 1

Model	Accuracy Score
KNN	0.8875
Logistic Regression	0.8903
Decision Tree	0.9159
Random Forest	0.9212
SVC	0.9055
Naive Bayes	0.7302

The table IV shows the results of 5-fold cross-validation on Dataset 2. The average accuracy scores achieved are: KNN (0.8962), Logistic Regression (0.9171), Decision Tree (0.9733), Naive Bayes (0.9328), SVC (0.9386), and Random Forest (0.9658). Decision Tree had the best accuracy, just followed by Random Forest; KNN that worst performance.

TABLE IV
5-FOLD CROSS VALIDATION ACCURACY SCORES FOR DATASET 2

Model	Accuracy Score
KNN	0.8962
Logistic Regression	0.9171
Decision Tree	0.9733
Random Forest	0.9658
SVC	0.9386
Naive Bayes	0.9328

IV. RESULTS AND ANALYSIS

• Result and Analysis for Dataset 1

The results in table V illustrated Logistic Regression, KNN, Random Forest, SVC, Decision Tree and Naive Bayes performance metrics showing accuracy, precision, recall, and F1-score metrics,

TABLE V
CLASSIFICATION MODELS PERFORMANCE COMPARISON

Performance	KNN	Logistic Regression	Decision Tree	Random Forest	SVC	Naive Bayes
Accuracy	0.8918	0.8974	0.9182	0.9258	0.9086	0.7328
Precision	0.89	0.90	0.92	0.92	0.91	0.88
Recall	0.89	0.90	0.92	0.93	0.91	0.73
F1 Score	0.89	0.90	0.92	0.92	0.91	0.76

Table VI showing training and testing accuracy metrics for the different models used.

TABLE VI
STATISTICS OF FORECASTING MODELS ACCURACY AND ERROR

Performance	Training	Testing
KNN	0.9252	0.8918
Logistic Regression	0.8907	0.8974
Decision Tree	0.9286	0.9182
Random Forest	0.9676	0.9258
SVC	0.9143	0.9086
Naive Bayes	0.7295	0.7328

According to the comparison of Tables V and VI, Random Forest performed the best overall, achieving a noteworthy high recall of 0.986 along with testing accuracy of 0.9258, precision of 0.92, recall of 0.93, and F1 score of 0.92. KNN achieved the lowest accuracy at 0.8918, while logistic regression produced findings that were balanced (accuracy 0.8974, F1 score 0.90). All models, with the exception of Naive Bayes (0.7328), had accuracy levels above 0.89; SVC (0.9086) and Decision Tree (0.9182) also did well. Fig. 8 provides additional illustrations of this conclusion.

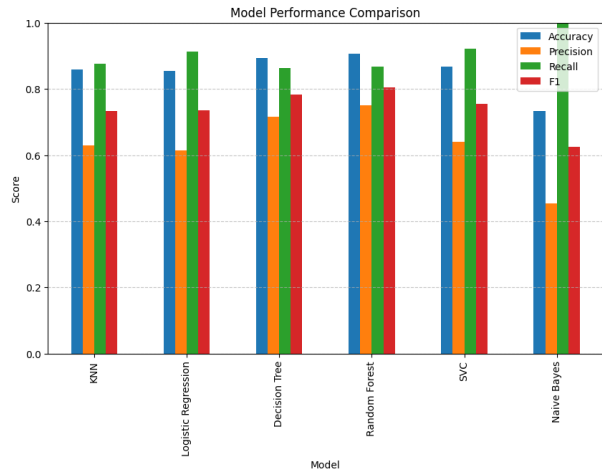


Fig. 8. First Dataset Model Performance Comparison

• Model Comparison in Training and Testing

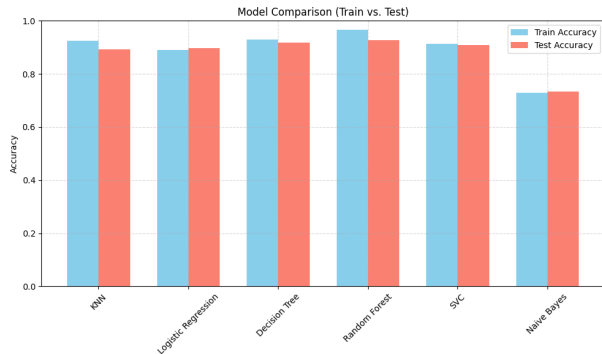


Fig. 9. Training vs Testing

As shown in Fig. 9 Random Forest outperformed the other models in the comparison, with 96.76% training and 92.58% testing accuracy.

ROC curves for first dataset as showing in Fig. '10 showing the performance of different classification models. Random Forest achieves the highest AUC (0.97), indicating strong ability to distinguish among the models

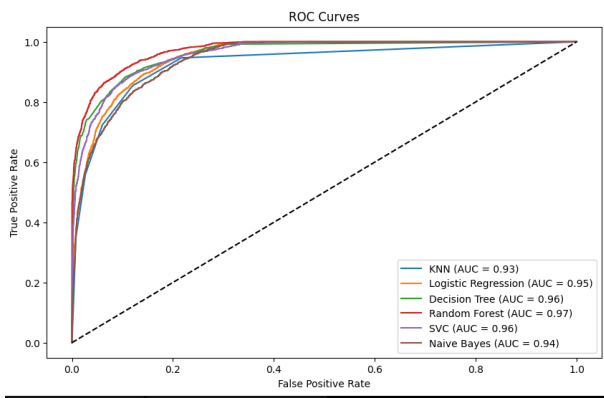


Fig. 10. Roc curves for dataset 2

• Result and Analysis for Dataset 2

The results in table VII demonstrated Logistic Regression, KNN, Random Forest, SVC, and Decision Tree performance metrics showing accuracy, precision, recall, and F1-score metrics,

TABLE VII
CLASSIFICATION MODELS PERFORMANCE COMPARISON

Performance	KNN	Logistic Regression	Decision Tree	Random Forest	SVC	Naive Bayes
Accuracy	0.8852	0.9297	0.9820	0.9649	0.9461	0.9563
Precision	0.89	0.93	0.98	0.97	0.95	0.96
Recall	0.89	0.93	0.98	0.96	0.95	0.96
F1 Score	0.89	0.93	0.98	0.97	0.95	0.96

TABLE VIII
TRAINING AND TESTING FOR DATASET 2

Performance	Training	Testing
KNN	0.9454	0.8852
Logistic Regression	0.9223	0.9297
Decision Tree	0.9847	0.9820
Random Forest	0.9739	0.9649
SVC	0.9570	0.9461
Naive Bayes	0.9505	0.9563

After looking to to tables VII and VIII and by visual comparison in Fig 12 it is clear that Decision Tree did the best. It had a high score overall with (0.9820) accuracy, in Precision 0.98 and recall of 0.98, showing strong skill in sorting. Random Forest was top in finding right cases with a recall of 0.96 KNN received the lowest scores, while logistic regression provided the most balanced results across all measures (accuracy: 0.9297, F1: 0.93) alongside Naive Bayes (0.956) and SVC (0.946) producing results that are similar.

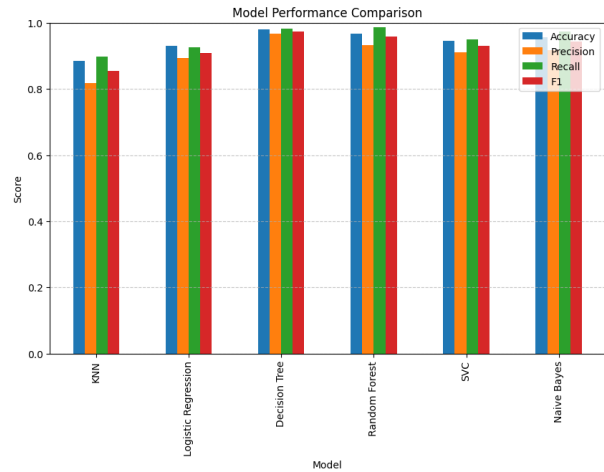


Fig. 11. Second Dataset Model Performance Compression

Model Comparison in Training and Testing:

Decision Tree outperformed the other models in the comparison, with 98.47% training and 98.20% testing accuracy. Compared to training accuracy (92.23%), logistic regression performed better in testing (92.97%), but Naive Bayes was stable (95.05% vs. 95.63%). While testing results were lower (88.52%), KNN had the highest training accuracy (94.54%). Above 94% accuracy, Random Forest and SVC both had excellent testing performance.

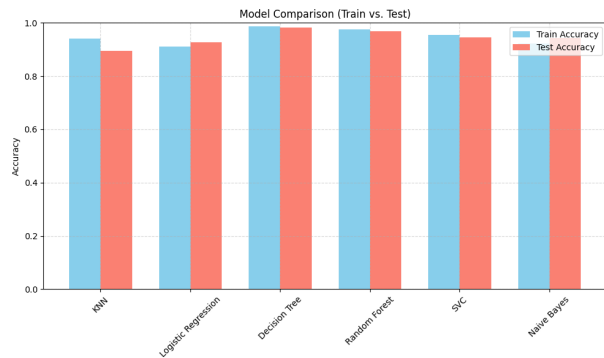


Fig. 12. Training vs testing

• ROC Curve

V. CONCLUSION

Using two datasets, this study assessed several machine learning models for predicting loan approval. With Random Forest and Decision Tree attaining around 92% and 91% accuracy on the first dataset, and 96% and 97% on the second, tree-based models consistently outperformed other models. KNN and Naive Bayes, on the other hand, performed worse, indicating their shortcomings when managing intricate feature interactions. Overall, the findings support the reliability of tree-based and ensemble methods for structured financial data.

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